Supplement

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Data-Generating Process

The effect of the treatment allocation variable A on Y is confounded with X because the covariate distributions in the treatment groups are different, like in an observational study.

Figure 1 shows the causal structure of the correlated covariates X_1, X_2, X_3, X_4, treatment allocation variable A, and survival time T. X_1 influences both A and T, X_2 only A, X_3 has no direct effect on either A or T but is highly correlated with the other covariates, and finally, X_4 influences only T.

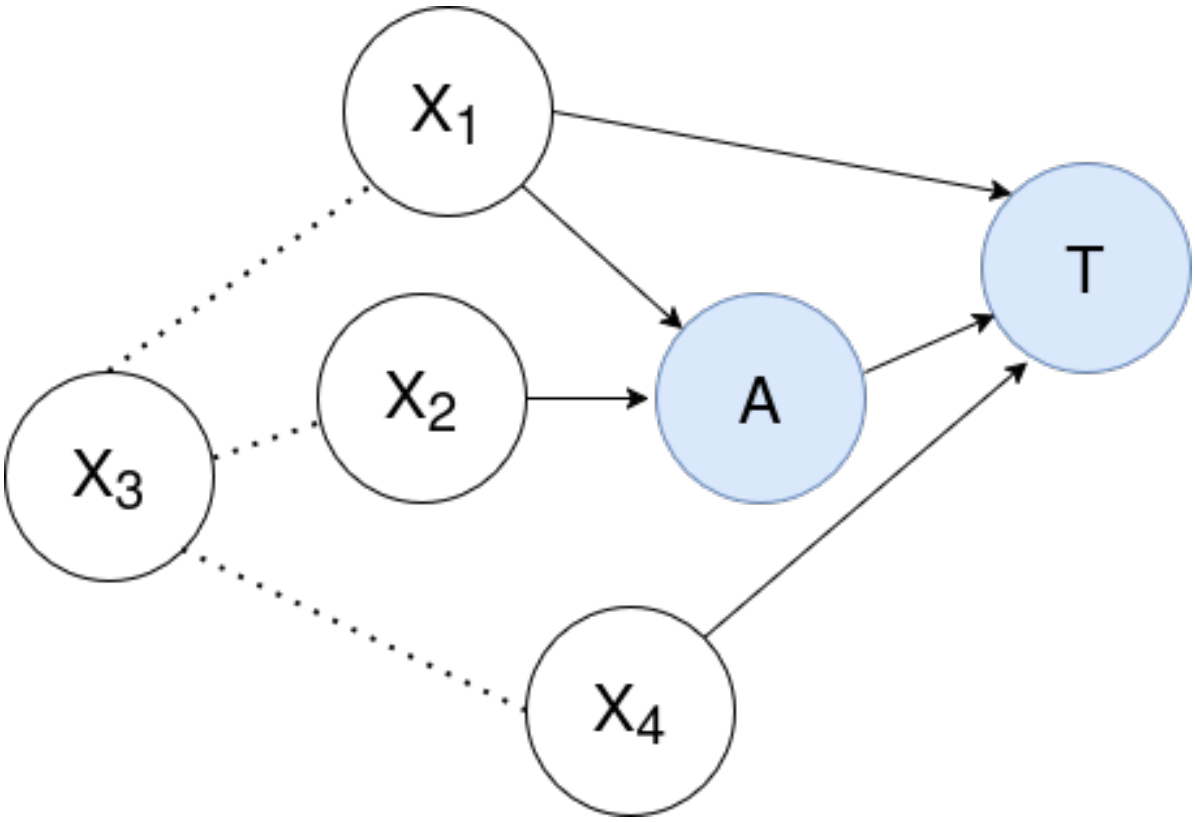


Figure 1: A graph visualizing the relationship between the variables X_1, X_2, X_3, X_4, treatment allocation A and survival time T.

Let $X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{pmatrix} \sim N_4(0, \Sigma)$, where each individual variance corresponding to the diagonal elements of Σ is 1 and the covariances are given by these pairwise correlations:

$$\begin{aligned} &\text{cor}(X_1, X_2) = 0.1, && \text{cor}(X_1, X_3) = 0.5, && \text{cor}(X_1, X_4) = 0.1, \\ &\text{cor}(X_2, X_3) = 0.6, && \text{cor}(X_2, X_4) = -0.2, && \text{cor}(X_3, X_4) = -0.5. \end{aligned}$$

The treatment assignment is denoted as $A \in \{0, 1\}$ and had the following probability:

$$\mathbb{P}(A = 1 | X_1, X_2) = logit^{-1}(-0.5 + 0.5X_1 + 0.8X_2) \tag{1}$$

This means that higher values of X_1 and X_2 correspond to a higher probability of receiving the treatment. This becomes apparent in Figure 2, where the treatment group has higher median

values for X_1, X_2 and also X_3 because of it's correlation structure. Since X_4 is negatively correlated with X_2, X_3 and only has a slightly positive correlation with X_1, the median of X_4 in the treatment group is slightly lower than in the control group.

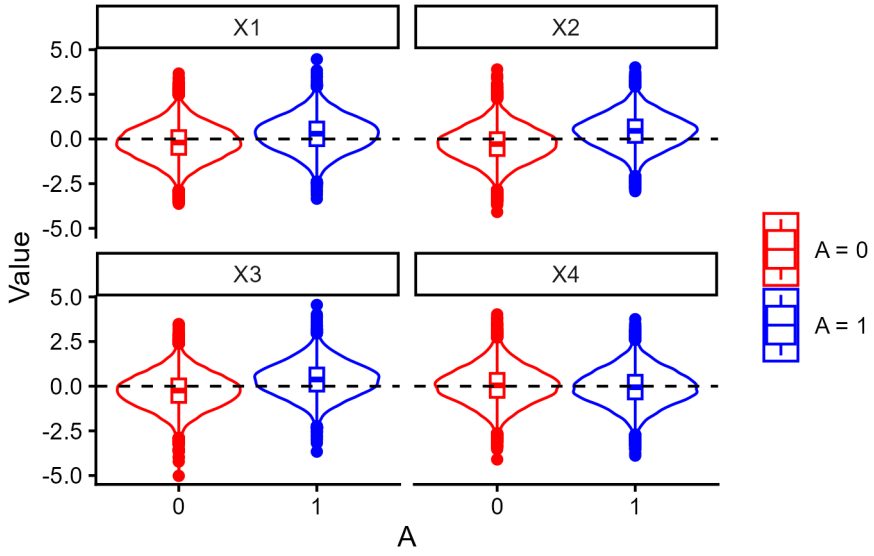


Figure 2: A plot showing the distribution of X_1, X_2, X_3, X_4 between control and treatment groups.

Survival times were generated from a Weibull distribution with:

- shape parameter: $k_i = 1 + \gamma \cdot A_i$, where $\gamma = 0$ corresponds to proportional hazards.
- scale parameter: $\lambda_i = \exp(\beta_1 X_{i1} + \beta_4 X_{i4} + \beta_5 \cdot A_i)$,

Then survival times are given by:

$$T_i = \left(\frac{-\log(U_i)}{\lambda_0 \cdot \lambda_i} \right)^{1/k_i}, \quad U_i \sim \text{Uniform}(0, 1) \tag{2}$$

where $\lambda_0 = 0.05$ is a constant multiplied with the scale parameter λ_i.

The censoring time was drawn from a mixture distribution:

$$C_i = 0.7 \cdot U_i + 0.3 \cdot \min(E_i, 30) \tag{3}$$

where $U_i \sim \text{Uniform}(0, c_{\max})$, $E_i \sim \text{Exponential}(\lambda_C)$. Taking $0.3 \cdot min(E_i, 30)$ ensures that no value greater than 9 can be contributed from the exponential distribution. The distribution

of censoring times is found further down in Figure 4. It should be noted that this form of censoring is *entirely independent* of covariate values.

Observed time and event indicator (i.e. status) are defined as

$$\tilde{T}_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i \leq C_i)$$

which produced a dataset that contains:

$$\{\tilde{T}_i, \delta_i, X_{i1}, X_{i2}, X_{i3}, X_{i4}, A_i\}_{i=1}^n$$

Scenarios

Based on the covariate adjusted difference in $RMST(\tau = 10)$ between the control and treatment group (i.e. the ATE for $RMST(\tau = 10)$), *four populations* with *three different amounts of censoring* were simulated, for a total of *12 scenarios*.

The four populations were simulated by four pairwise parameter sets: for each population $p \in \{1, 2, 3, 4\}$, the parameters were assigned as $\beta_5 = \{-0.5, 0, 0.5, -1\}_p$ and $\gamma = \{0, 0, -0.3, 0.49\}_p$. That is, the p-th element from each vector was used together.

As seen in Figure 3, Population 1 shows a clear positive ATE (ATE=0.777, i.e. group with $A = 1$ has higher RMST) with proportional hazards. Population 2 has the exact same survival curves with proportional hazards and thus no ATE at all (ATE=0). Population 3 corresponds to no difference in the ATE (ATE=0, raw ATE = 0.399) even though the survival curves are crossing (=non-proportional hazards). Population 4 shows a mild positive ATE (ATE=0.224, raw ATE = 0.528) with non-proportional hazards and crossing survival curves. Descriptive and True were calculated as written in the ATE section of the main paper.

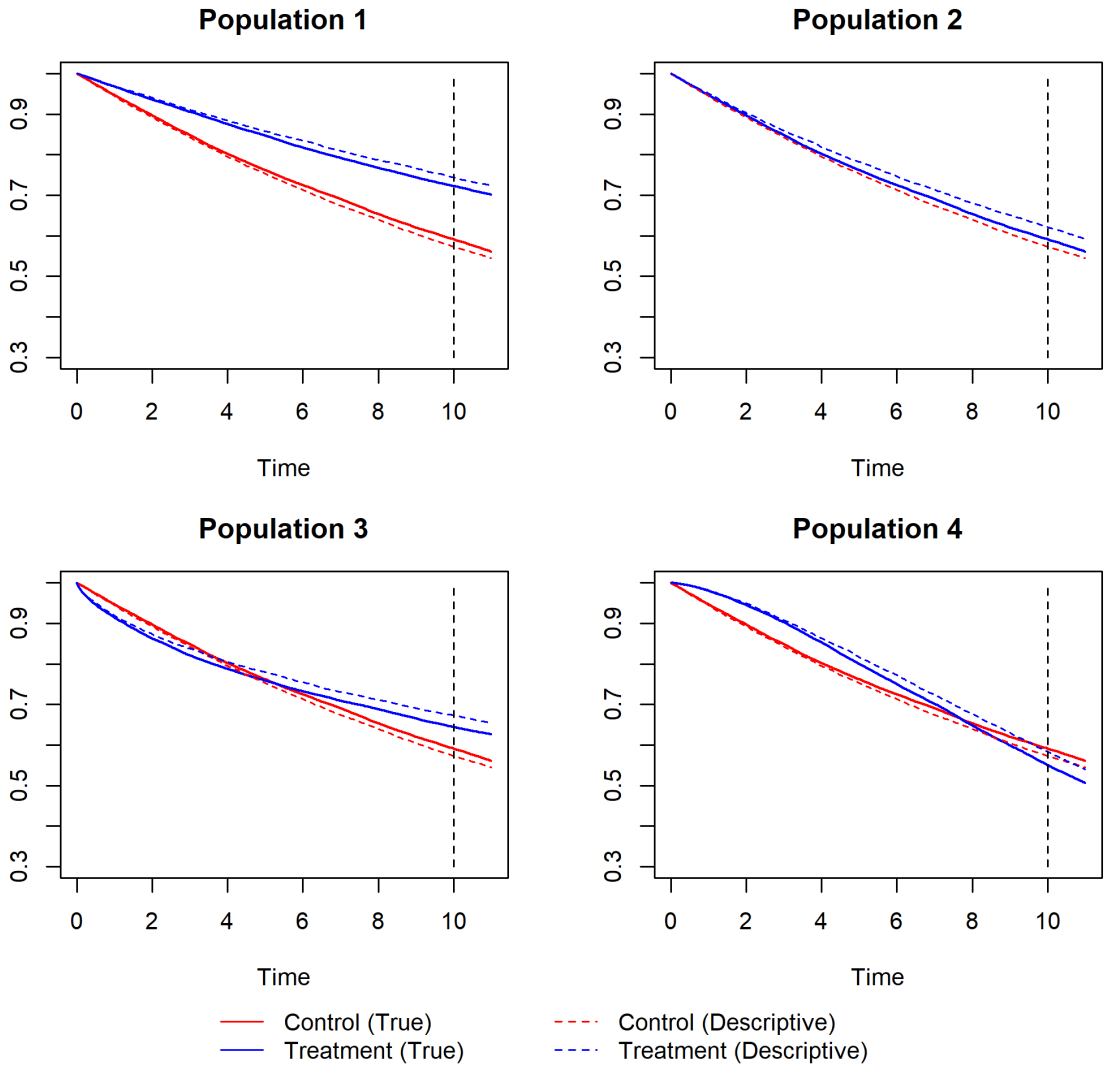


Figure 3: Survival functions ("True") and Kaplan-Meier estimated survival curves ("Descriptive", unadjusted for X) of the treatment groups of all populations.

20% censoring was reached by setting $c_{max} = 20$ and $\lambda_C = 0.02$. 40% was reached with $c_{max} = 20$ and $\lambda_C = 0.06$. Finally, 60% censoring was reached by $c_{max} = 15$ and $\lambda_C = 0.08$. Figure 4 shows the cumulative distribution of censoring times when these values are inserted into the mixture distribution of equation 3.

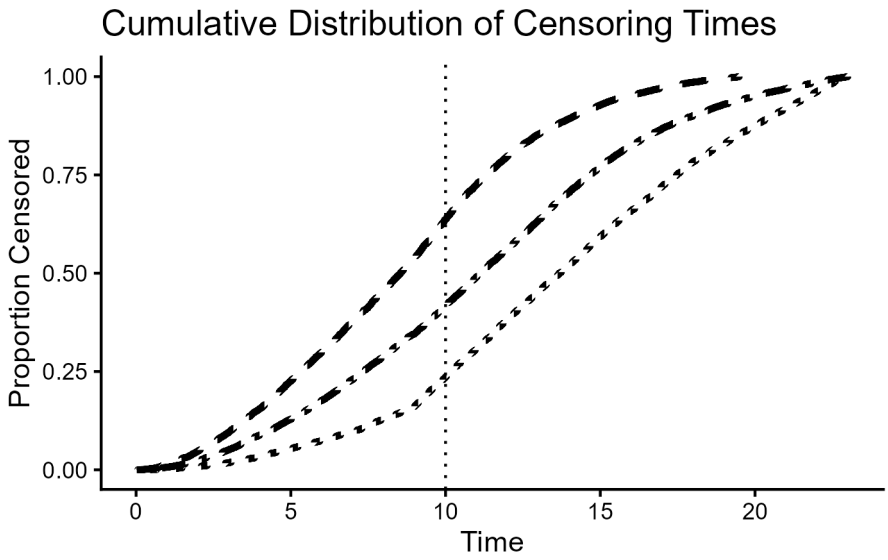


Figure 4: Cumulative Distribution of censoring times, reaching 20%, 40% and 60% censoring at $\tau = 10$.

Confidence Bands of RCS Smoother

Figure 5, Figure 6, Figure 7 and Figure 8 show the RCS smoother confidence bands of the method with the generally lowest (RF) and highest ($GEE_{complex}$) variance.

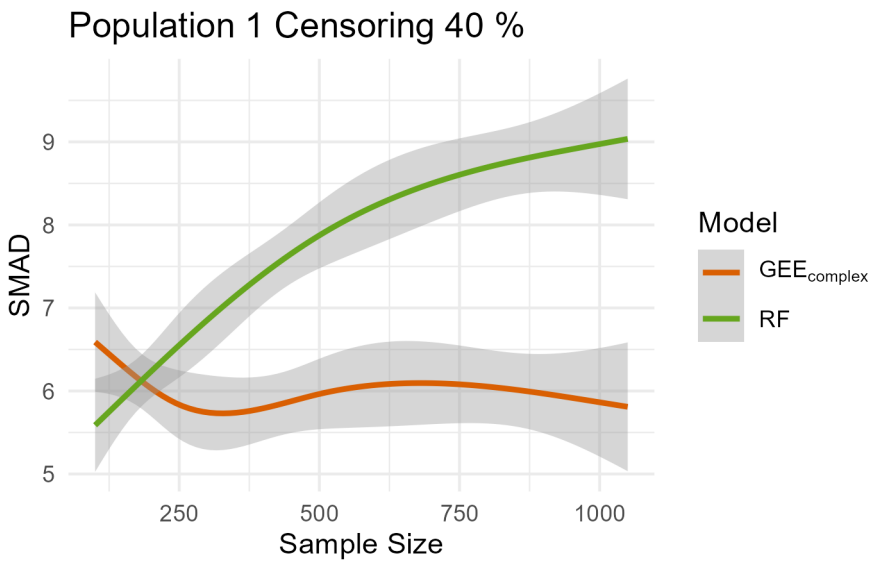


Figure 5: {A plot showing the SMAD of RF and $GEE_{complex}$ with confidence bands for the RCS smoother.

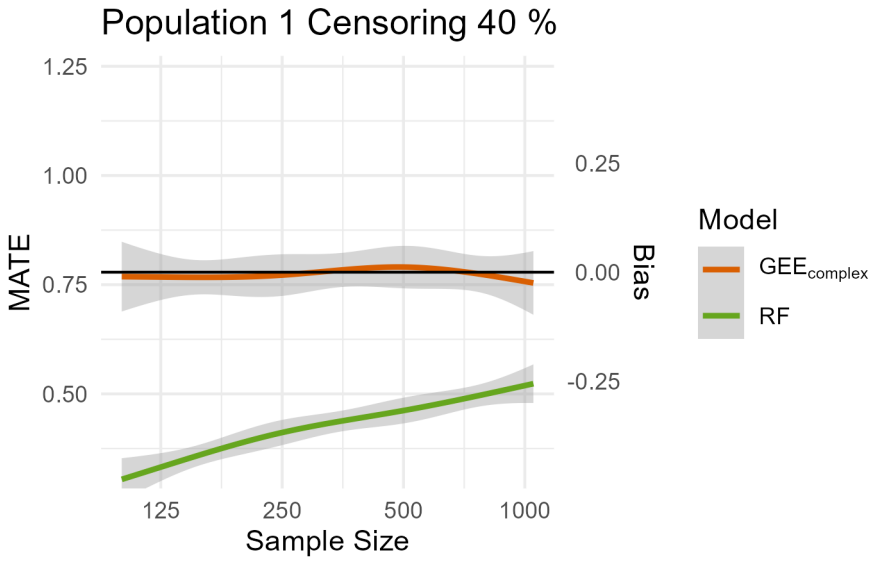


Figure 6: {A plot showing the MATE of RF and $GEE_{complex}$ with confidence bands for the RCS smoother.

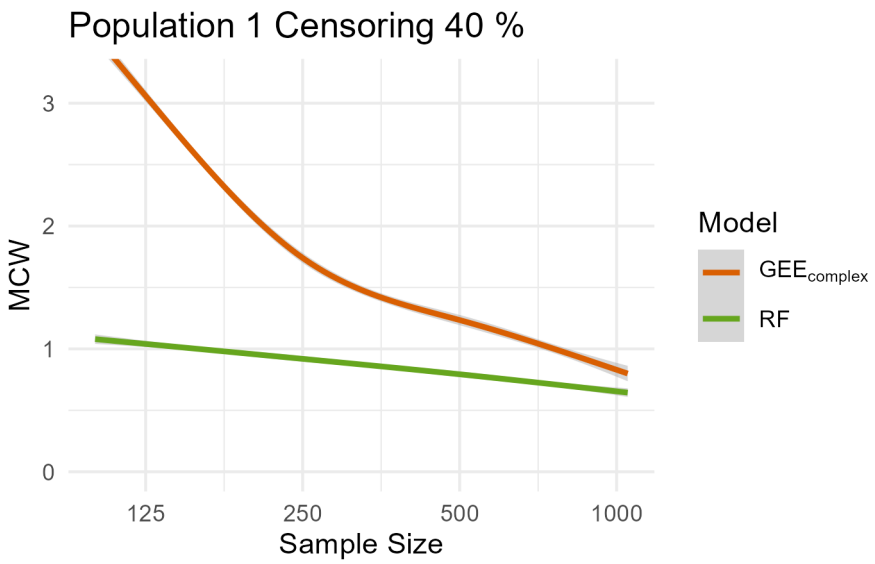


Figure 7: {A plot showing the MCW of RF and $GEE_{complex}$ with confidence bands for the RCS smoother.

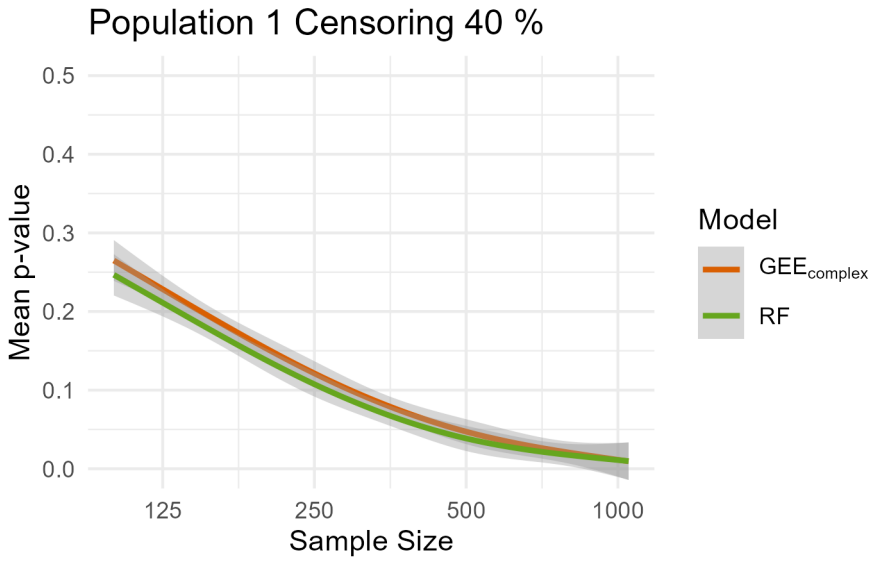


Figure 8: {A plot showing the Mean p-value of RF and $GEE_{complex}$ with confidence bands for the RCS smoother.

Additional Results

Results for 20% and 60% censoring level are found here. **SMAD**: Figure 9, Figure 10. **MATE**: Figure 11, Figure 12. **MCW**: Figure 13, Figure 14. **Mean p-value**: Figure 15, Figure 16.

SMAD

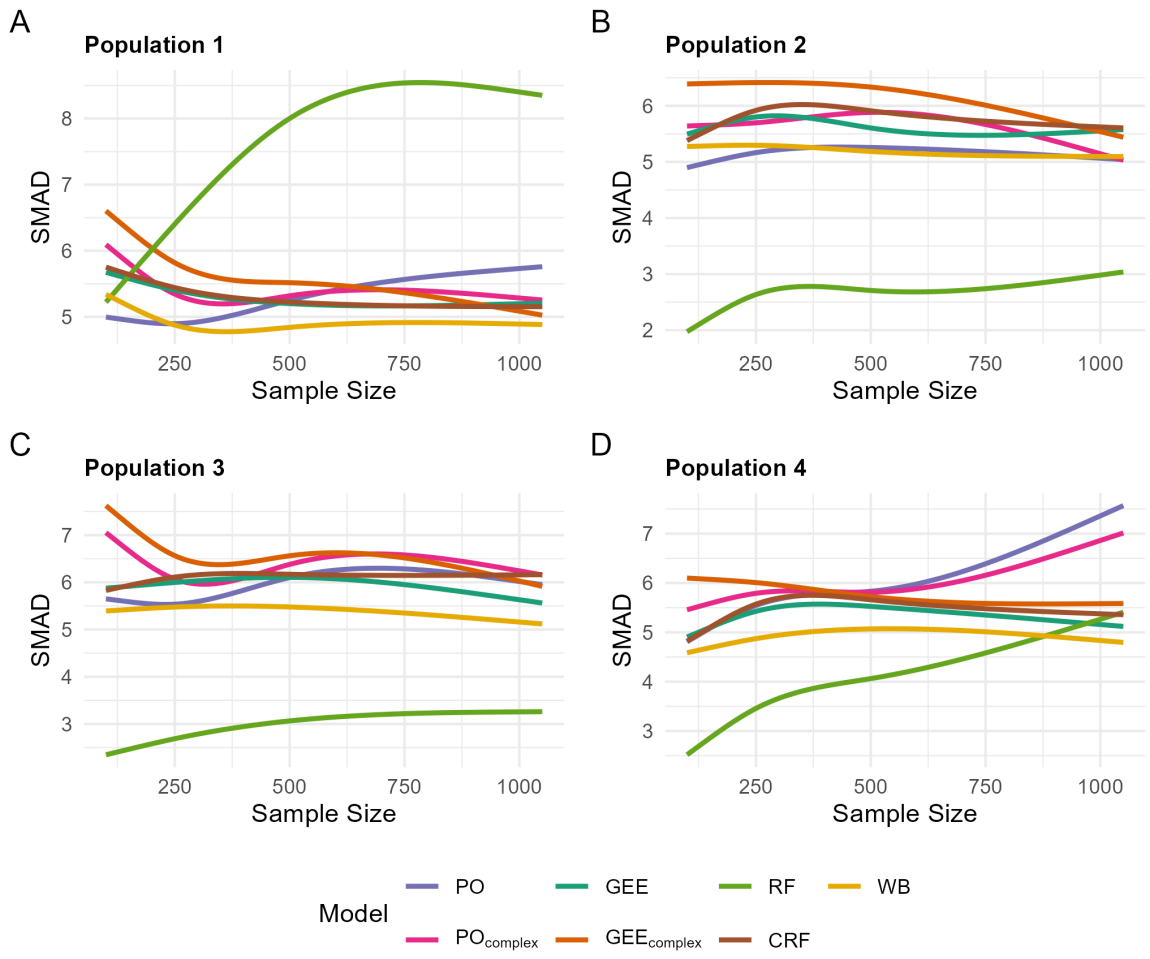


Figure 9: RCS smoothed curves of SMAD for all populations with 20% censoring.

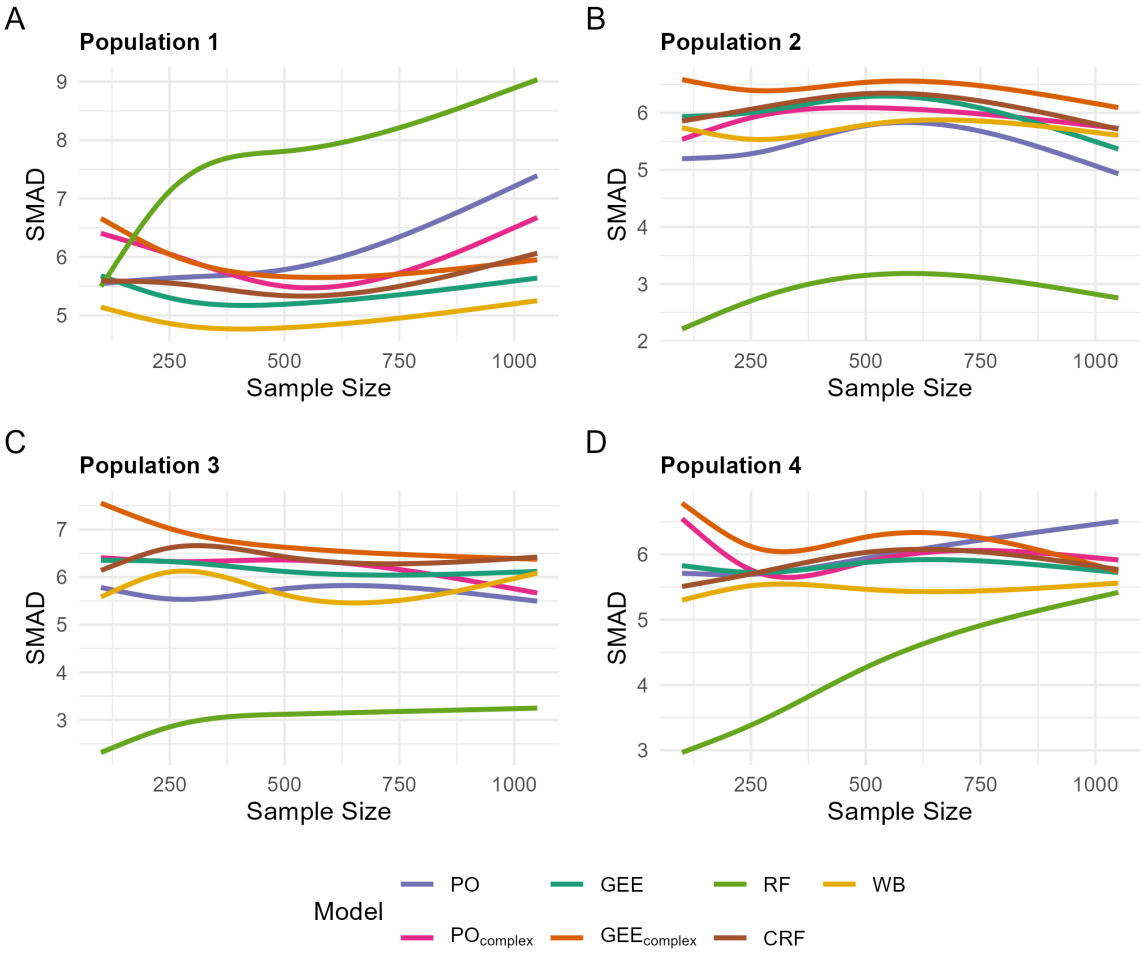


Figure 10: RCS smoothed curves of SMAD for all populations with 60% censoring.

MATE

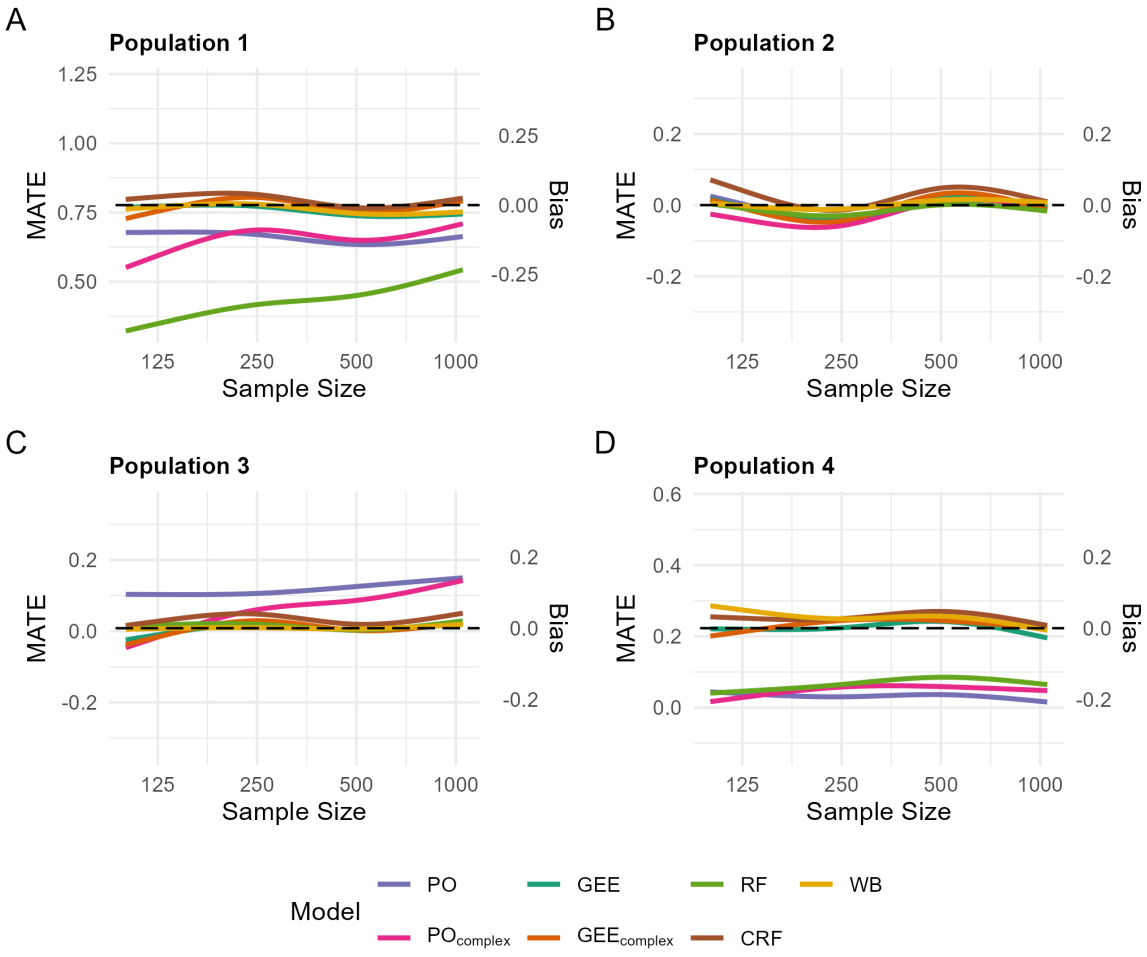


Figure 11: RCS smoothed curves of MATE for all populations with 20% censoring.

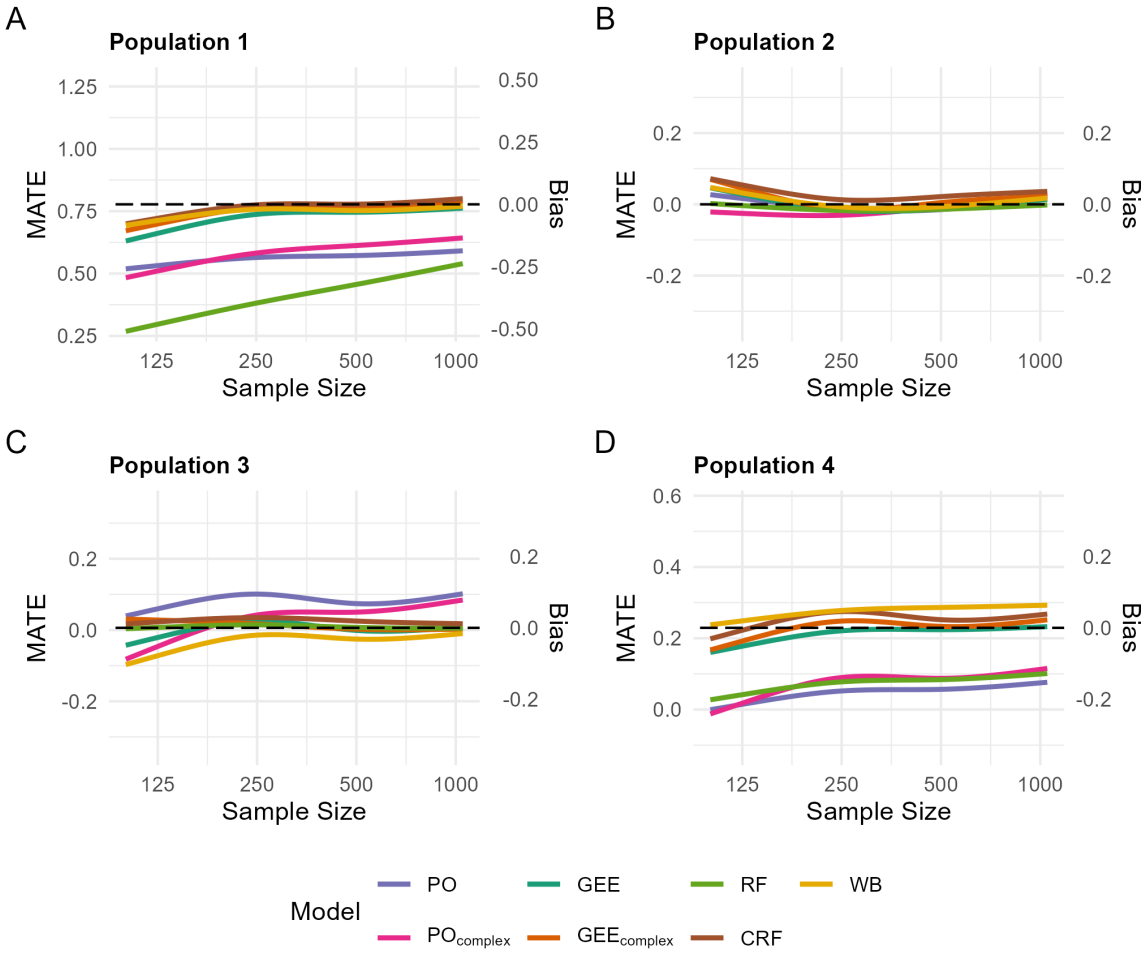


Figure 12: RCS smoothed curves of MATE for all populations with 60% censoring.

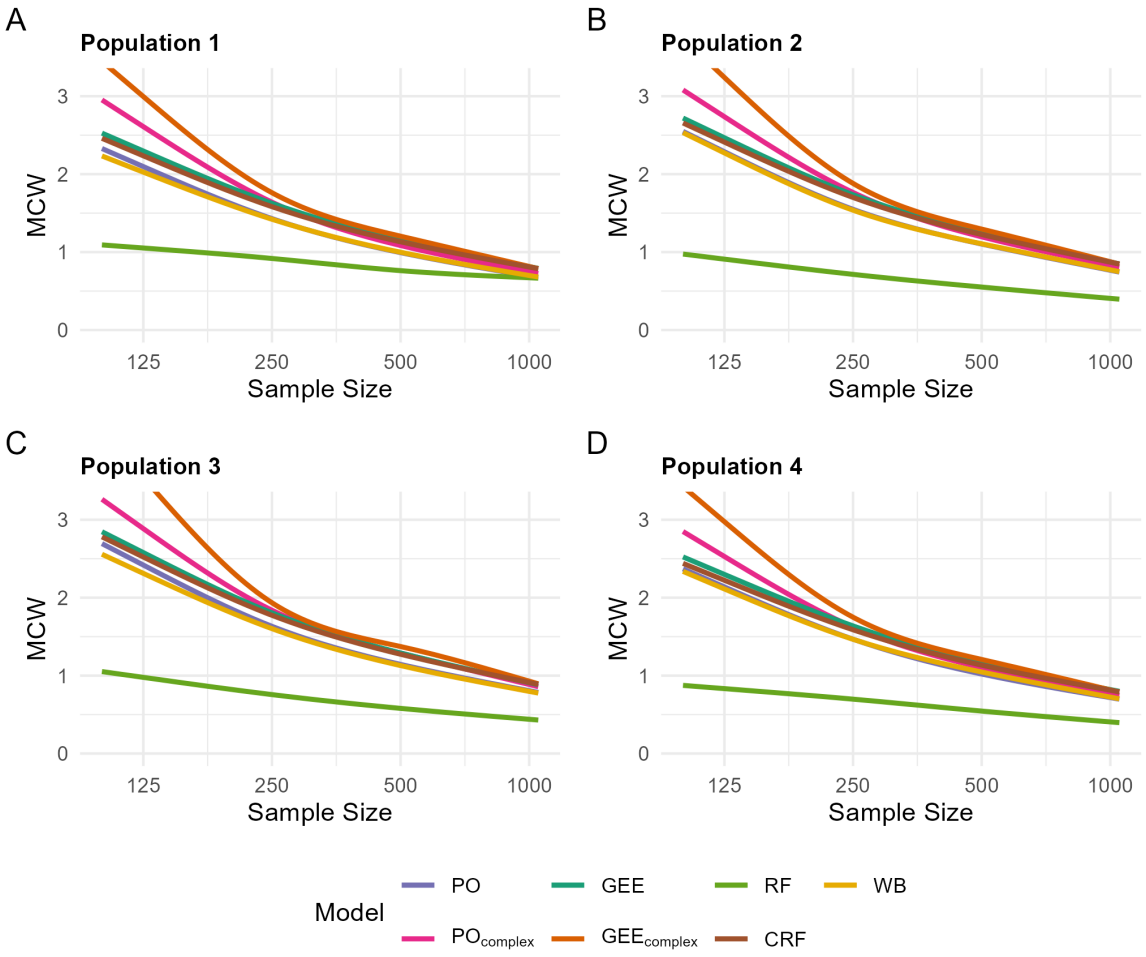


Figure 13: RCS smoothed curves of MCW for all populations with 20% censoring.

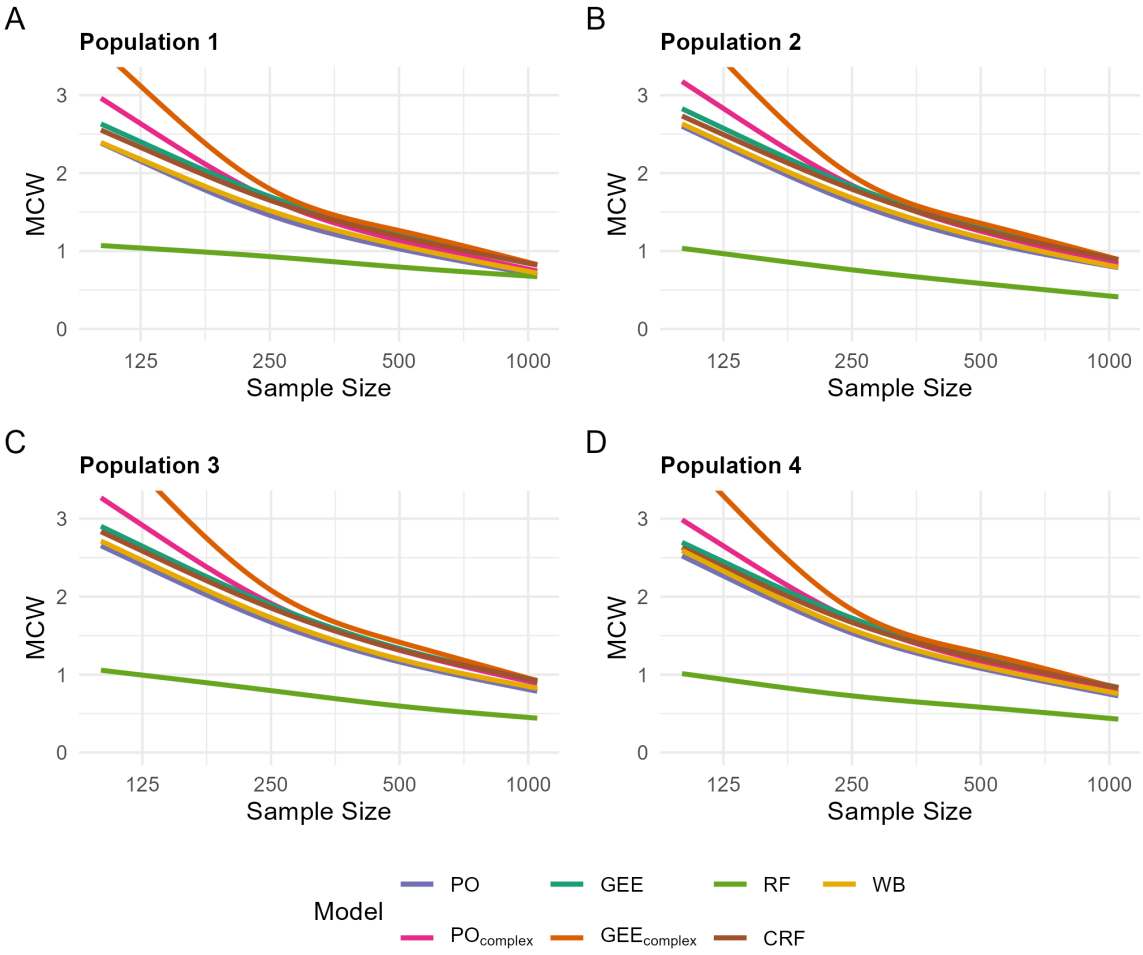


Figure 14: RCS smoothed curves of MCW for all populations with 60% censoring.

Mean p-value

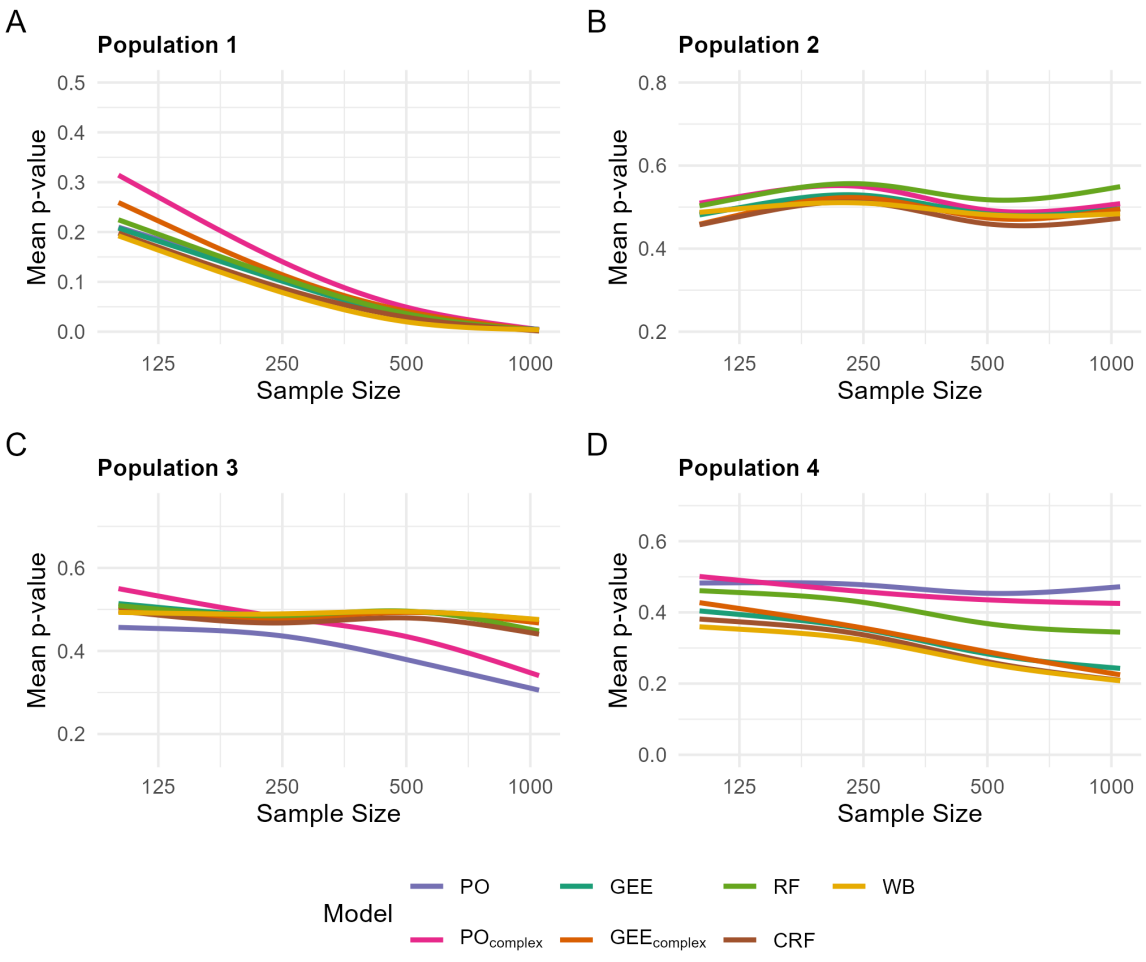


Figure 15: RCS smoothed curves of Mean p-value for all populations with 20% censoring.

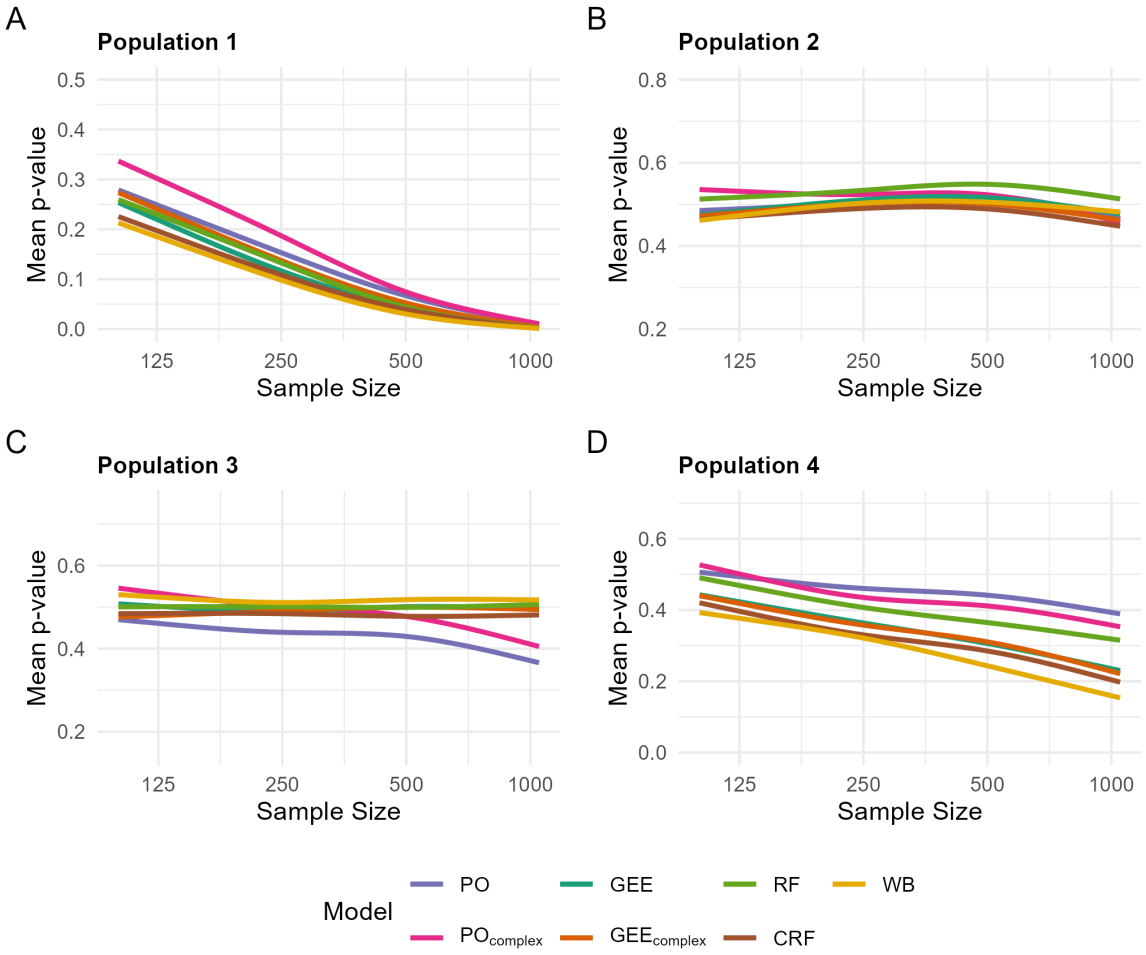


Figure 16: RCS smoothed curves of Mean p-value for all populations with 60% censoring.